

DigitVision AI

Intelligent Handwritten Digit Recognition using Machine Learning and
Artificial Neural Networks

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Google Colab Notebook: DigitVision AI - Google Colab Notebook

Streamlit Application: DigitVision AI - Streamlit Application

1 Abstract

Handwritten digit recognition is one of the most fundamental applications of Artificial Intelligence and Computer Vision. It enables computers to automatically recognize handwritten numerical digits and has numerous applications in postal automation, bank cheque processing, document digitization, educational systems, and optical character recognition (OCR).

This project presents **DigitVision AI**, an intelligent handwritten digit recognition system developed using an Artificial Neural Network (ANN) trained on the MNIST handwritten digit dataset. The dataset contains 60,000 training images and 10,000 testing images representing handwritten digits from 0 to 9. Before training, the images were normalized and converted into one-dimensional feature vectors consisting of 784 input features.

The proposed ANN architecture consists of two fully connected hidden layers with ReLU activation, Dropout regularization, and a Softmax output layer for multi-class classification. The trained model achieved a test accuracy of **98.40%**, demonstrating excellent performance on unseen handwritten digits.

To make the model interactive and user-friendly, the trained ANN was deployed using Streamlit Community Cloud. The application enables users to draw handwritten digits on a digital canvas and obtain real-time predictions together with confidence scores and processed image visualization.

2 Introduction

Handwritten digit recognition is a classical image classification problem that has significantly contributed to the advancement of Artificial Intelligence and Machine Learning.

The objective is to automatically recognize handwritten numerical digits and classify them into one of the ten digit classes ranging from 0 to 9.

Automatic digit recognition plays an important role in postal automation, bank cheque verification, document digitization, educational systems, and Optical Character Recognition (OCR). Since handwriting styles vary considerably among individuals, traditional image processing techniques often struggle to achieve high recognition accuracy.

Artificial Neural Networks (ANNs) overcome this limitation by learning complex feature representations directly from large datasets. In this project, the MNIST handwritten digit dataset is used to train an ANN model developed using TensorFlow and Keras. After image preprocessing and normalization, the trained model is evaluated using multiple performance metrics and deployed through Streamlit as an interactive web application for real-time handwritten digit prediction.

The developed system achieved an overall test accuracy of **98.40%**, demonstrating the effectiveness of Artificial Neural Networks for handwritten digit recognition while providing a simple and accessible user interface.

3 Objectives

The primary objectives of this project are:

- Develop an Artificial Neural Network (ANN) for handwritten digit recognition.
- Train and evaluate the model using the MNIST handwritten digit dataset.
- Perform image preprocessing through normalization and feature extraction.
- Evaluate model performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
- Deploy the trained model as an interactive Streamlit web application for real-time handwritten digit prediction.

4 Dataset

The project uses the **MNIST Handwritten Digit Dataset**, one of the most widely used benchmark datasets for handwritten digit recognition. The dataset contains grayscale images of handwritten digits from 0 to 9 and is commonly used for evaluating image classification models.

The dataset consists of 60,000 training images and 10,000 testing images. Each image has a resolution of 28×28 pixels and is normalized before training. The images are then flattened into one-dimensional feature vectors containing 784 input features.

Table 1: Dataset Summary

Parameter	Value
Dataset	MNIST Handwritten Digits
Training Images	60,000
Testing Images	10,000
Image Size	28×28 Pixels
Color Format	Grayscale
Number of Classes	10
Input Features	784

5 Methodology

The proposed system follows a standard deep learning pipeline for handwritten digit recognition. The workflow begins by loading the MNIST handwritten digit dataset. The images are then preprocessed through normalization and flattening before being supplied to the Artificial Neural Network.

The ANN model is trained using TensorFlow and Keras with the Adam optimizer and Sparse Categorical Crossentropy loss function. Early Stopping and Model Checkpoint callbacks are used to improve generalization and preserve the best-performing model.

Finally, the trained model is evaluated using standard classification metrics and deployed as a Streamlit web application for real-time handwritten digit prediction.

6 ANN Architecture

The proposed Artificial Neural Network consists of two fully connected hidden layers followed by Dropout layers for regularization. The final output layer uses the Softmax activation function to classify handwritten digits into one of the ten output classes.

Table 2: ANN Model Architecture

Layer	Configuration
Input Layer	784 Neurons
Hidden Layer 1	Dense (256), ReLU
Dropout	0.30
Hidden Layer 2	Dense (128), ReLU
Dropout	0.20
Output Layer	Dense (10), Softmax

The network contains a total of **235,146** trainable parameters. During training, the model achieved a test accuracy of **98.40%**, demonstrating excellent performance in recognizing handwritten digits.

7 Results and Discussion

The performance of the proposed Artificial Neural Network (ANN) was evaluated using standard classification metrics including Accuracy, Precision, Recall, and F1-Score. The model demonstrated excellent performance on the MNIST handwritten digit dataset. Table 3 summarizes the obtained results.

Table 3: Model Performance Summary

Metric	Accuracy	Precision	Recall	F1-Score
ANN Model	98.40%	98.40%	98.40%	98.40%

The ANN model achieved an overall test accuracy of **98.40%**. The high Precision, Recall, and F1-Score indicate that the model effectively learned handwritten digit patterns while maintaining excellent generalization on unseen test samples.

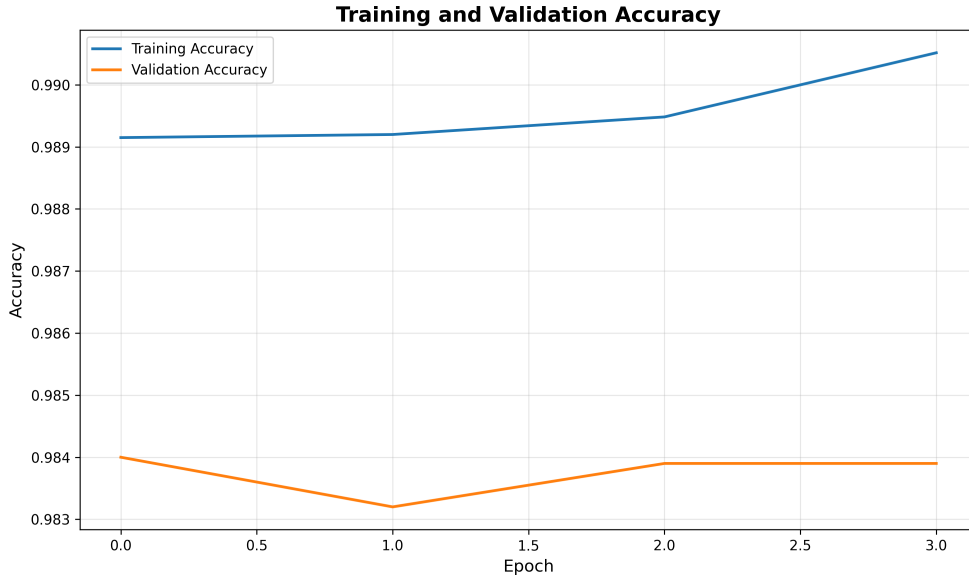


Figure 1: Training and Validation Accuracy

Figure 1 illustrates the variation in training and validation accuracy during model training. The model converged rapidly and maintained consistently high validation accuracy, demonstrating stable learning throughout the training process.

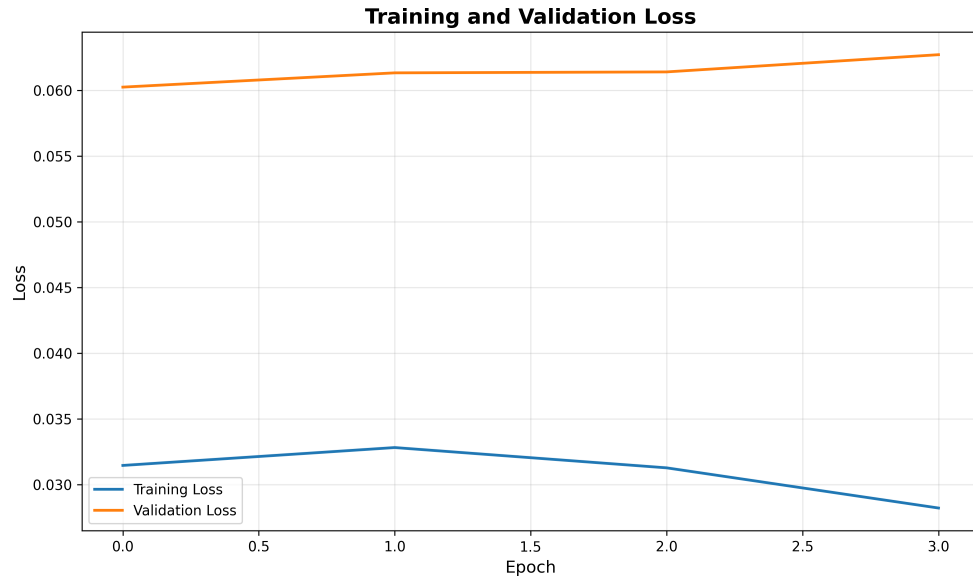


Figure 2: Training and Validation Loss

Figure 2 shows the corresponding training and validation loss. The decreasing loss values indicate effective optimization with minimal overfitting due to the use of Dropout regularization and Early Stopping.

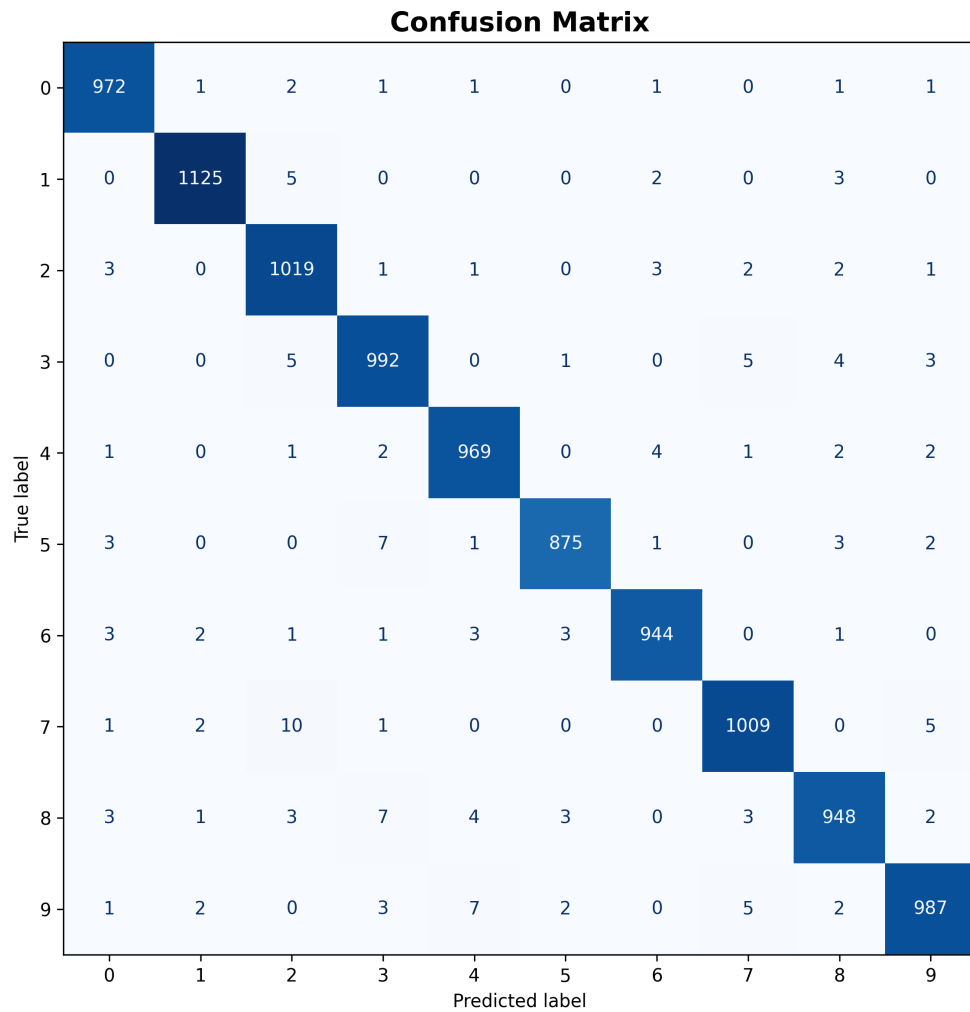


Figure 3: Confusion Matrix

Figure 3 presents the confusion matrix of the trained ANN model. Most handwritten digits were classified correctly, with only a few visually similar digits being misclassified, resulting in an overall accuracy of 98.40%.

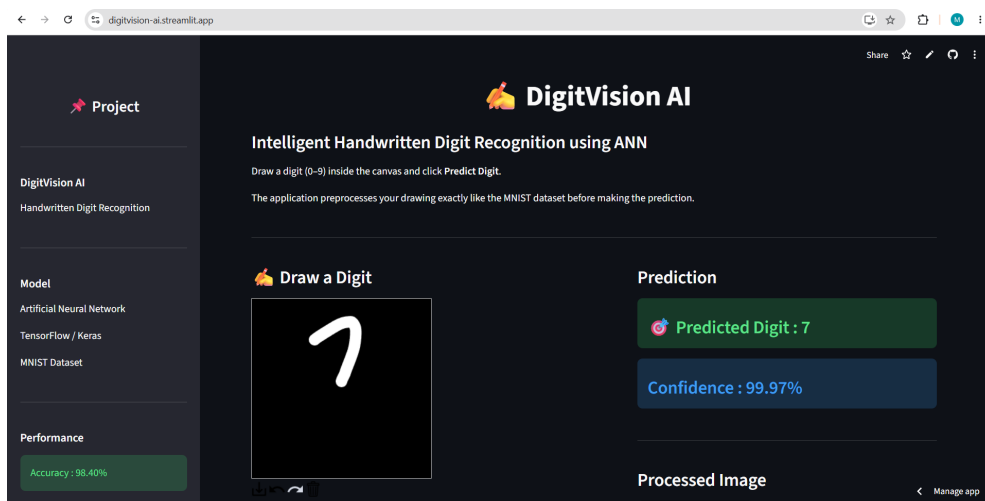


Figure 4: DigitVision AI Streamlit Web Application

Figure 4 shows the deployed Streamlit web application. The application enables users to draw handwritten digits, perform real-time predictions, visualize the processed image, and view confidence scores through a simple and interactive interface.

8 Conclusion

This project presented an end-to-end handwritten digit recognition system using Artificial Neural Networks. The MNIST handwritten digit dataset was preprocessed through normalization and feature extraction before training the ANN model using TensorFlow and Keras.

Experimental evaluation showed that the proposed ANN achieved an overall test accuracy of **98.40%**, demonstrating excellent classification performance. The trained model was successfully deployed as a Streamlit web application, allowing users to perform real-time handwritten digit prediction through an intuitive web interface.

9 Future Scope

The proposed system can be further enhanced by:

- Implementing Convolutional Neural Networks (CNNs) for improved recognition accuracy.
- Supporting recognition of multiple handwritten digits simultaneously.
- Integrating webcam-based real-time handwritten digit recognition.
- Deploying the trained model as a REST API.
- Developing Android and iOS applications for mobile-based digit recognition.

10 References

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5. MNIST Handwritten Digit Database. <http://yann.lecun.com/exdb/mnist/>